

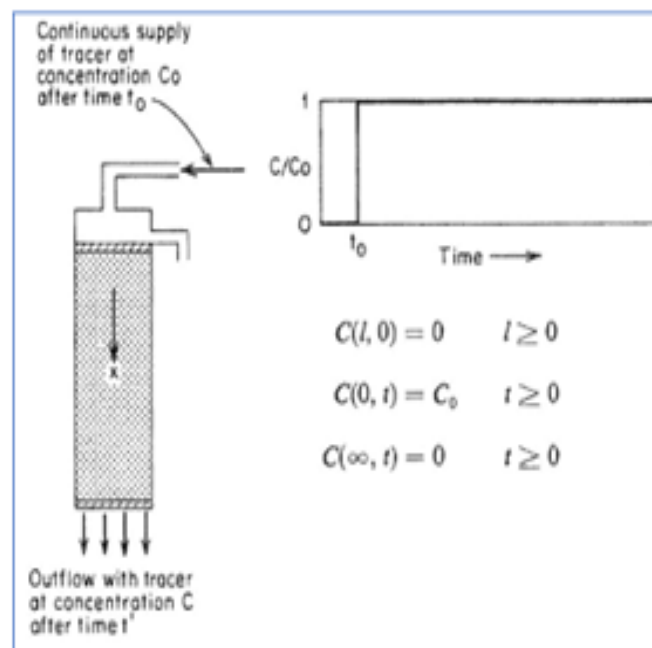
ES 668 Environmental Computational Laboratory
Spring 2025

Lab Assignment-4 (Solving PDEs with MATLAB)

DEMO PROBLEM

1. The 1-D solute transport of a conservative substance injected with a step function input to a saturated porous media (homogeneous and isotropic) can be modelled with the governing PDE as follows:

$$D_l \frac{\partial^2 C}{\partial l^2} - \bar{v}_l \frac{\partial C}{\partial l} = \frac{\partial C}{\partial t}$$



Solve the PDE using 'pdepe' solver in MATLAB with the following data provided:

Length of column (L) = 2.5 m

Average linear velocity (v) = 1 m/s

Coefficient of hydrodynamic dispersion (D) = 0.01 m²/s

Input boundary concentration (C_0) = 1 kg/m³

Meshing to be used for numerical solution: time steps: 100 ; distance intervals = 100.

- (a) Plot the surface plot for variation of concentration with 't' and 'x'.
- (b) Plot the concentration vs time plot at the outlet end of the column.
- (c) Plot the concentration front in the column at any appropriate time step.

Answer: Script - `function` PDESol

```
%parameters
T=5;
L=2.5;
D=0.01;
v=1;
ui=1;
u0=0;
%time and distance span
x=linspace(0,L,100);
t=linspace(0,T,100);
%solve PDE
sol=pdepe(0,@pbpde,@icpb,@bcpb,x,t);
u_output=sol(:,end);
u_middle=sol(40,:);
%Plot results
surf(x,t,sol)
title('pollutant concentration over space and time')
xlabel('Distance(m)')
ylabel('Time(s)')
zlabel('Concentration(x,t)')
grid on;
figure;
plot(t,u_output);
xlabel('Time(s)')
ylabel('Concentration(kg/m^3)')
grid on;
figure;
plot(x, u_middle);
xlabel('Distance (m)')
ylabel('Concentration (kg/m^3)')
grid on;

function [c,f,s]=pbpde(x,t,u,dudx);
c=1;
f=D*dudx;
s=-v*dudx;
end
%initial condition
function u0=icpb(x);
u0=0;
end
%boundary condition
function [p1,q1,pr,qr]=bcpb(xl,ul,pr,qr,t);
p1=ul-ui;
q1=0;
pr=0;
qr=1;
end
end
```

Graph-

pollutant concentration over space and time

